

# Essential Linux Command Cheat Sheet

**Introduction:** Your foundational toolkit  for navigating, managing, and interacting with files and directories in the Linux command line – crucial for DevOps!

---

## Block 1: Fundamentals & Navigation

- **Basic Syntax:**
    - command [options] [inputs]
    - (Visual Idea: A simple flowchart  : [Inputs]  [ Command + Options ]  [ Output / Action ])
  - **User Prompts:** Know your power level!
    - \$ : Normal User  (Limited privileges)
    - # : Root/Admin/Super User  (Full privileges - use caution!)
  - **File System Hierarchy (Core Linux):** The "tree" of your system! 
    - (Visual Idea: A vertical tree diagram starting from the root)
  - / (Root Directory - The Boss! )
  - |— bin, sbin, etc, var, usr... (Important System Areas )
  - |— home (Where regular users live )
  - | |— ec2-user (Your User's Home - Your Room! )
  - | |— another\_user
  - |— root (Root User's Home - The Admin Office )
  - |— mnt, media (Where external things connect )
  - |— ... other directories
    - *Note:* /c/users/siva is typical in Windows environments using Linux tools (like WSL), linking to your Windows home.   
- **Current Location:**
  - /home/ec2-user: Example Linux user home
  - /c/users/siva: Example Windows user home (WSL)
  - /root: Root user home

- **Change Directory (cd):** Moving around the tree! 🚶
    - cd <directory>: Go directly to a specific branch.
    - cd ..: Go up one level (back towards the root). ⬆
    - cd or cd ~: Go straight to your home directory. 🏠
  - **Print Working Directory (pwd):** Where are you right now? 🤖
    - pwd: Display the full path of your current location. 📌
- 

## Block 2: Listing Directory Contents (ls)

- **ls:** See what's inside a directory! 📁 (Lists files/folders alphabetically)
  - **Options:** Customize your view! 👁
    - -l: Long listing format (Permissions, owner, size, date, name). (Visual Idea: Show a sample output line with columns labeled: Permissions | Owner | Size | Date | Name)
    - -lr: Long, **Reverse** alphabetical order. ⬇ A-Z
    - -lt: Long, Sorted by modification **T**ime (Newest files on top). 🕒 ⬇ (New to Old)
    - -ltr: Long, Sorted by modification **T**ime (Oldest files at bottom). 🕒 ⬆ (Old to New)
    - -la: Long, List **A**ll files (including hidden files starting with '.'). 👻
- 

## Block 3: File & Directory Management (Creation, Deletion, Copy, Move)

- **Create File:**
  - touch <file-name>: Creates an empty file. 📄 ✨
- **Create Directory:**
  - mkdir <name>: Creates a new directory (folder). 📁 +
- **Remove Directory:**
  - rmdir <name>: Removes an **empty** directory. 📁 ✖ (Only works if it's empty!)
- **Remove Files & Directories:** Delete with caution!
  - rm <file-name>: Removes a file. (May ask "Are you sure?")

- `rm -f <file-name>`: Forcefully removes a file (No asking!). 💥
  - `rm -rf <directory-name>`: Recursively and Forcefully deletes a directory and **all** its contents. (THE DANGER COMMAND! 🔥 💀 Use only if you are 100% sure!)
  - **Copy:** Duplicating things. 📄 ➡ 📄
    - `cp <source> <destination>`: Copies files.
    - `cp -r <source_dir> <destination_dir>`: Copies directories recursively (needed for folders). 📁 ➡ 📁
  - **Move/Rename:** Cutting or renaming. ✂ ➡ 📄
    - `mv <source> <destination>`: Moves files/directories. Also used to rename files: `mv <old_name> <new_name>`.
- 

## Block 4: File Content & Redirection

- **View File Content:**
    - `cat <file-name>`: Display the whole file content on screen. 📖 👁
  - **Create/Edit File Content (with Redirection >):** Writing to files! 🖋
    - `cat > <file-name>`: Type text, press Enter, then Ctrl+D to save. **REPLACES** any previous content! ⚠
    - `cat >> <file-name>`: Type text, press Enter, then Ctrl+D to save. **APPENDS** text to the end. ✨
  - **Redirection (> & >>):** Sending output elsewhere. ➡
    - (Visual Idea: [Command Output] ➡ > (Overwrite!) ➡ [ File ])
    - (Visual Idea: [Command Output] ➡ >> (Append!) ➡ [ File ])
- 

## Block 5: Searching, Filtering & Monitoring

- **Search within Files (grep):** Find lines with specific text. 🔍 📄
  - `grep <word-to-search> <file>`: Finds lines matching the pattern in a file.
  - `command | grep <word-to-search>`: **Piping!** Send output from one command as input to grep.
    - (Visual Idea: [Command] ➡ | (Pipe) ➡ [ grep ] ➡ [ Filtered Output ])

- **Find Files (find):** Search for files by name or other criteria. 
    - find <where to search> -name <filename>: Search for a file name starting from a path.
  - **Monitor Log Files (tail):** Watch files grow! 
    - tail -f <log-file>: Shows the end of a file and displays new lines as they are added (like watching a live feed!). 
  - **awk Command:** (Advanced text processing and reporting - mention it exists for complex pattern matching and data extraction). 
- 

## Block 6: Text Editor (vim)

- **Command:** vim <file-name>: Open or create a file in the visually improved editor. 
- **3 MODES:** The core concept of Vim! You need to know which mode you are in! 
  - (Visual Idea: 3 interconnected bubbles or boxes with clear labels and arrows showing transitions)
  - **1. Normal / Esc Mode:** (The starting mode. For navigation, commands, copy/paste/delete).
    - Enter from anywhere: Press Esc. 
    - u: **U**ndo last change. 
    - yy: **Y**ank (copy) the current line. 
    - p: **P**aste below the current line. 
    - dd: **D**elete (cut) the current line. 
    - gg: Go to the **g**o top of the file. 
    - Shift+g (or G): Go to the **g**o bottom of the file. 
  - **2. Insert Mode:** (For typing actual text into the file). 
    - Enter from Normal: Press i (insert before cursor), a (append after cursor), o (open new line below), etc. (Show i as the main entry point).  Insert Mode
    - Exit to Normal: Press Esc. Insert Mode  Normal

- **3. Command-Line / Colon Mode:** (For actions like saving, quitting, searching, replacing). 🗨️ 💾 🔍
  - Enter from Normal: Press :. (The cursor goes to the bottom left).  
Normal ➡️ Command-Line
  - Exit: After command runs or press Esc. Command-Line ➡️ Normal
  - :q: Quit (if no changes). 🙌
  - :wq: **W**rite (save) and **q**uit. ✅ 💾 🙌
  - :wq!: Force **w**rite and **q**uit. (Ignores warnings, be careful!) ⚡ 💾 🙌
  - :q!: **Q**uit without saving (discard changes). ❌ 💾
  - :/<pattern>: **S**earch **f**orward for the pattern. 🔍 ⬇️
  - :?<pattern>: **S**earch **b**ackward for the pattern. 🔍 ⬆️
  - :noh: **N**o highlight (clear search highlighting). ✨
  - :<line\_num>d: **d**elete specific line number (e.g., :27d). ✂️ #27
  - :%d: **d**elete **a**ll lines (% means whole file). ⚡ 📄
  - :<line\_num>s/<old>/<new>/: On <line\_num>, substitute (replace) the **f**irst <old> with <new>. ↻
  - :<line\_num>s/<old>/<new>/g: On <line\_num>, substitute (replace) **a**ll <old> with <new>. ↻ ↻ ↻
  - :%s/<old>/<new>/g: In the **e**ntire **f**ile, substitute (replace) **a**ll <old> with <new>. ↻ 🌐 ↻

---

**Conclusion:** These commands are building blocks! Keep practicing in your terminal or a sandbox environment. Happy commanding! 👍 😊

**Suraj Kumar Padhi.**